TechSent: Sentiment Analysis of Tweets on Tech Products

Social Media Sentiment Intelligence

A Multi-Model Analysis of Apple vs Google Brand Perception
Lingua Franca Data Elites October
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1. Executive Summary

TechSent leverages social media text analytics and transformer-based modeling to assess the public perception of two global technology leaders: Apple and Google. The project quantifies sentiment patterns, identifies emotion drivers, and deploys a real-time Streamlit app for predictive insights.

Key Highlights:
- Fine-tuned DistilBERT model achieved 98% accuracy in sentiment classification.
- Apple tweets exhibited higher emotional engagement, both positive and negative.
- Google sentiment leaned neutral, reflecting product-centric rather than emotional discourse.
- Final Streamlit dashboard enables real-time brand sentiment monitoring.

Business Impact:
- Empowers marketing and PR teams to react dynamically to public sentiment.
- Provides data-driven insights into product perception and reputation management.
- Enables automated brand tracking and trend forecasting.

2. Project Background

Technology brands operate in an ecosystem where consumer sentiment directly affects market value. Public opinion shared on social media, especially Twitter, influences purchasing decisions, loyalty, and advocacy.

This project aimed to:
- Quantify and compare emotional sentiment between Apple and Google.
- Build a scalable pipeline for continuous sentiment monitoring.

- Translate NLP model outputs into strategic business insights.

Scope:
- Focused exclusively on Apple and Google mentions.
- Classified sentiment into Positive, Neutral, and Negative.
- Built on a labeled dataset from Data.World (CrowdFlower: Brands and Product Emotions).

Objective: Deliver a deployable sentiment intelligence framework for continuous brand analysis.

3. Methodology

Data Sources:
- CrowdFlower Dataset — 'Brand and Product Emotions' (9,000 tweets)

Data Cleaning & Standardization:
- Lowercased text and removed URLs, emojis, and hashtags.
- Filtered for Apple and Google mentions only.
- Removed duplicates, nulls, and irrelevant brand references.
- Balanced dataset via under sampling for class uniformity.

Feature Engineering:
- Tokenization and encoding using DistilBERT tokenizer (tok).
- Maximum sequence length: 128 tokens.
- Labels encoded: 0 = Negative, 1 = Neutral, 2 = Positive.

4. Exploratory Data Analysis (EDA)

Brand Distribution:
- Apple: 52% of total tweets
- Google: 48% of total tweets

Sentiment Distribution:
Positive: 35%
Neutral: 58%
Negative: 7%

Observation: Apple had stronger polar sentiment, where more users expressed excitement or frustration compared to Google.

Word Frequency:
- Positive Tweets: love, smooth, beautiful, update, battery life
- Negative Tweets: bug, crash, lag, expensive, poor support

Insight: Positive words associated with design and innovation (Apple); negative with performance or stability (Google).

5. Key Findings & Discoveries

- DistilBERT model captured complex contextual sentiment better than classical models.
- Apple had 15% higher positive engagement compared to Google.
- Google's neutrality reflects its utilitarian user base; less emotional but consistent.
- Common triggers for negativity were product updates, software glitches, and pricing concerns.

6. Technical Implementation

Tools & Libraries:
- Python (Jupyter Notebook)
- Libraries: transformers, torch, pandas, numpy, sklearn, matplotlib, streamlit

Techniques Used:
- Transformer fine-tuning with DistilBERT-base-uncased.
- Cross-entropy loss optimization using Adam.
- Stratified 80/20 data split.
- Evaluation Metrics: Accuracy, Precision, Recall, F1-score.

Model and tokenizer saved via: model.save_pretrained('sentiment_model')
tok.save_pretrained('sentiment_model')

7. Business Applications

- Real-time sentiment tracking for marketing and PR teams.
- Brand comparison analytics for competitive strategy.
- Customer experience monitoring post-product launches.
- Early warning system for negative publicity events.

8. Limitations

- Dataset limited to English-language tweets (Western sentiment bias).
- Snapshot dataset instead of live streaming data.
- Sentiment categories simplified to three (no emotion granularity).
- Twitter context (sarcasm, humor) may affect model precision.

9. Next Steps

- Integrate Twitter/X API for live data collection.

- Expand to multilingual sentiment for global insights.
- Deploy enhanced model with aspect-based sentiment analysis.
- Automate retraining pipeline using Hugging Face Hub + Airflow.

10. Final Recommendations

- Use Streamlit dashboard for real-time brand monitoring and reporting.
- Schedule model retraining every 3–6 months to capture evolving language trends.
- Combine sentiment output with sales and engagement metrics for richer insights.
- Integrate dashboard into brand management workflows for daily use.

11. Appendices

- Data Dictionary: Sentiment labels and variable descriptions.
- Model Details: Hyperparameters, sequence length, batch size. - Sample Predictions: Text - Sentiment - Confidence scores.
- Visuals: EDA plots, confusion matrix, ROC curves, word clouds.

Team Members

Huldah Rotich – Group Lead

Leon Wairachu

Maureen Ngaire

Teresia Ndung'u

Emmanuel Chol

Carlton Ogola

Github: https://github.com/Ngaire44/Lingua_Franca